Internship Report

Title: Twitter Analytics Dashboard Development using Power BI

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Internship Organization: ElevanceSkills Technology Private Limited

Internship Domain: Data Analytics

Duration: 1 Month

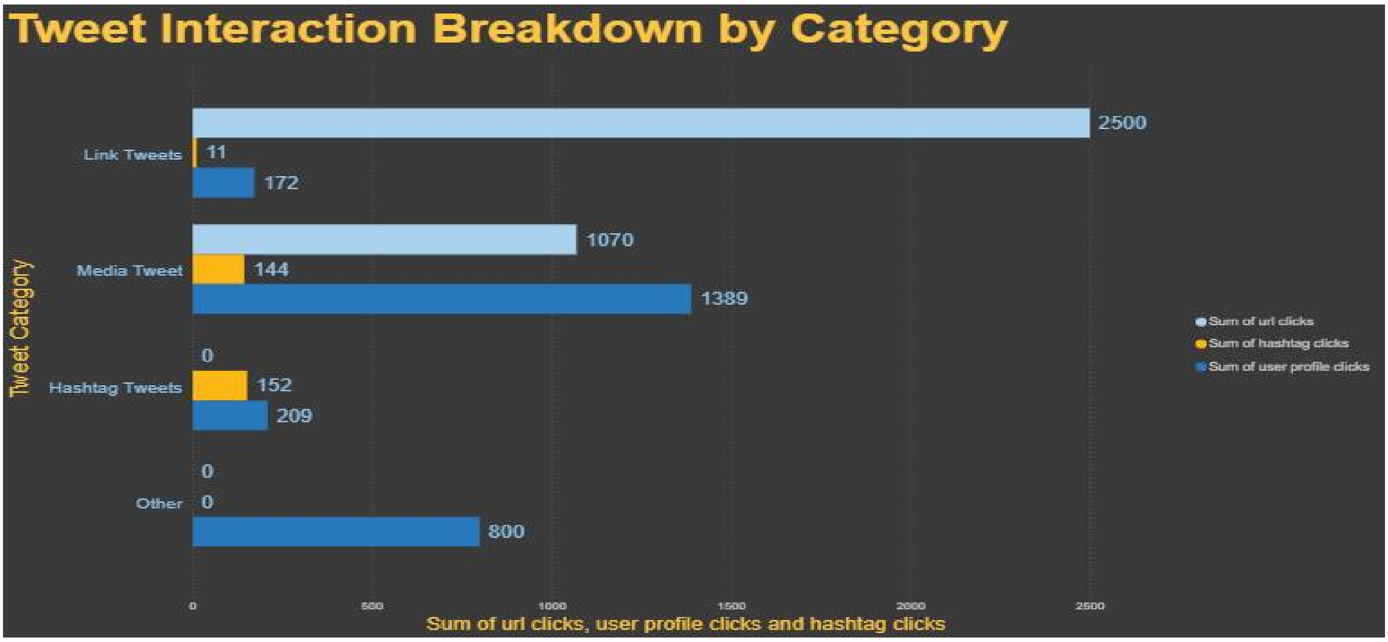
Introduction
The purpose of this internship was to develop practical skills in data analytics using Microsoft Power BI. Throughout the internship, I worked on analysing a Twitter analytics dataset and created multiple interactive dashboards based on different business requirements. The internship helped me understand how raw data can be transformed into meaningful insights using data visualization, DAX calculations, and filtering techniques.

Background
Social media platforms generate a large amount of user interaction data every day. Organizations use this information to understand audience behaviour, engagement, and content performance. During this internship, I analysed Twitter engagement data containing metrics such as likes, retweets, replies, media interactions, impressions, engagement rates, and user activities. Power BI was used to visualize this data and identify meaningful trends.

Learning Objectives
The primary objectives of this internship were:
- Learn data visualization using Microsoft Power BI.
- Understand data cleaning and transformation techniques.
- Apply DAX formulas for calculated columns and measures.
- Create interactive dashboards using different chart types.
- Analyze social media engagement metrics.
- Improve analytical thinking and dashboard design skills.

Activities and Tasks
During the internship, I completed the following analytical tasks:

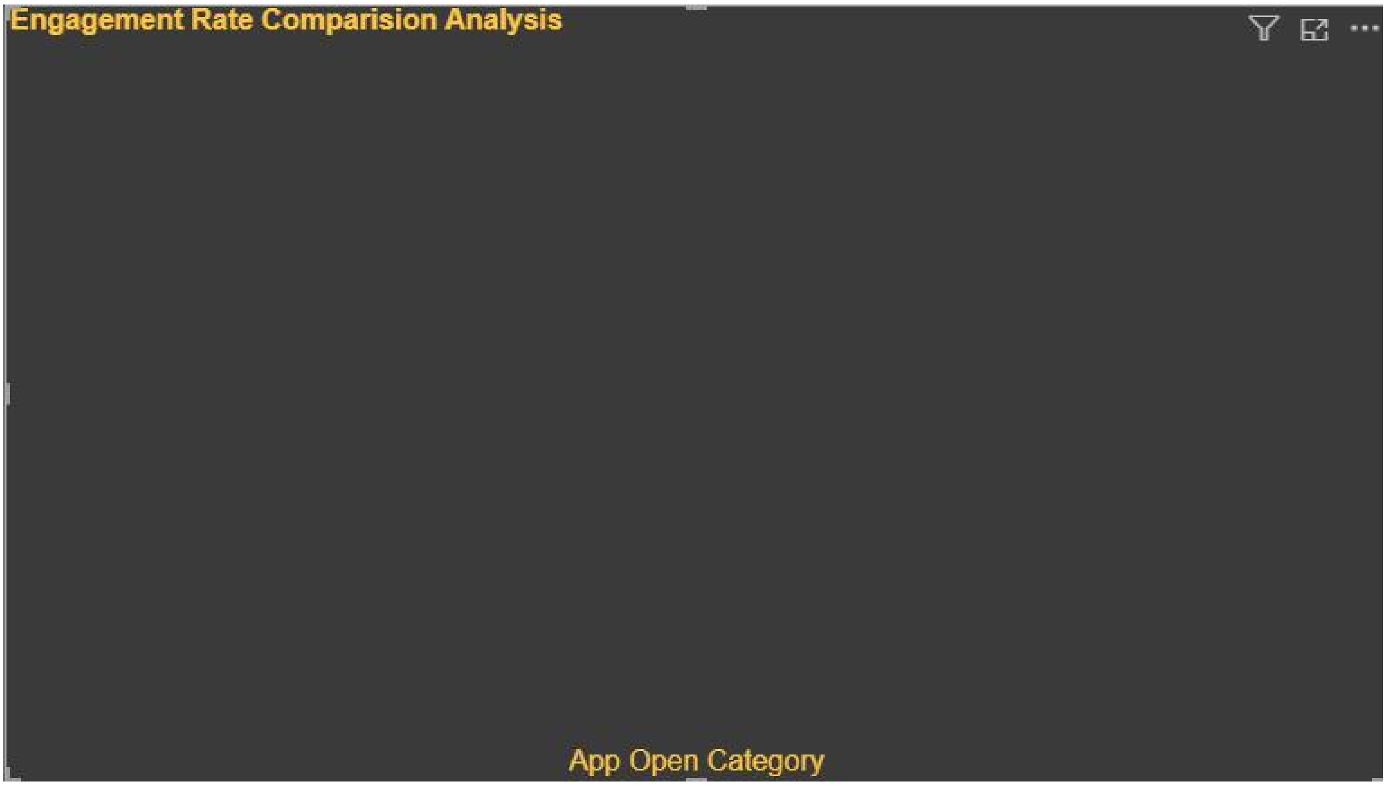
Task 1
Created a clustered bar chart to analyse URL clicks, user profile clicks, and hashtag clicks grouped by tweet categories. Applied filters based on interaction availability, tweet date, word count, and visibility conditions.



Task 2

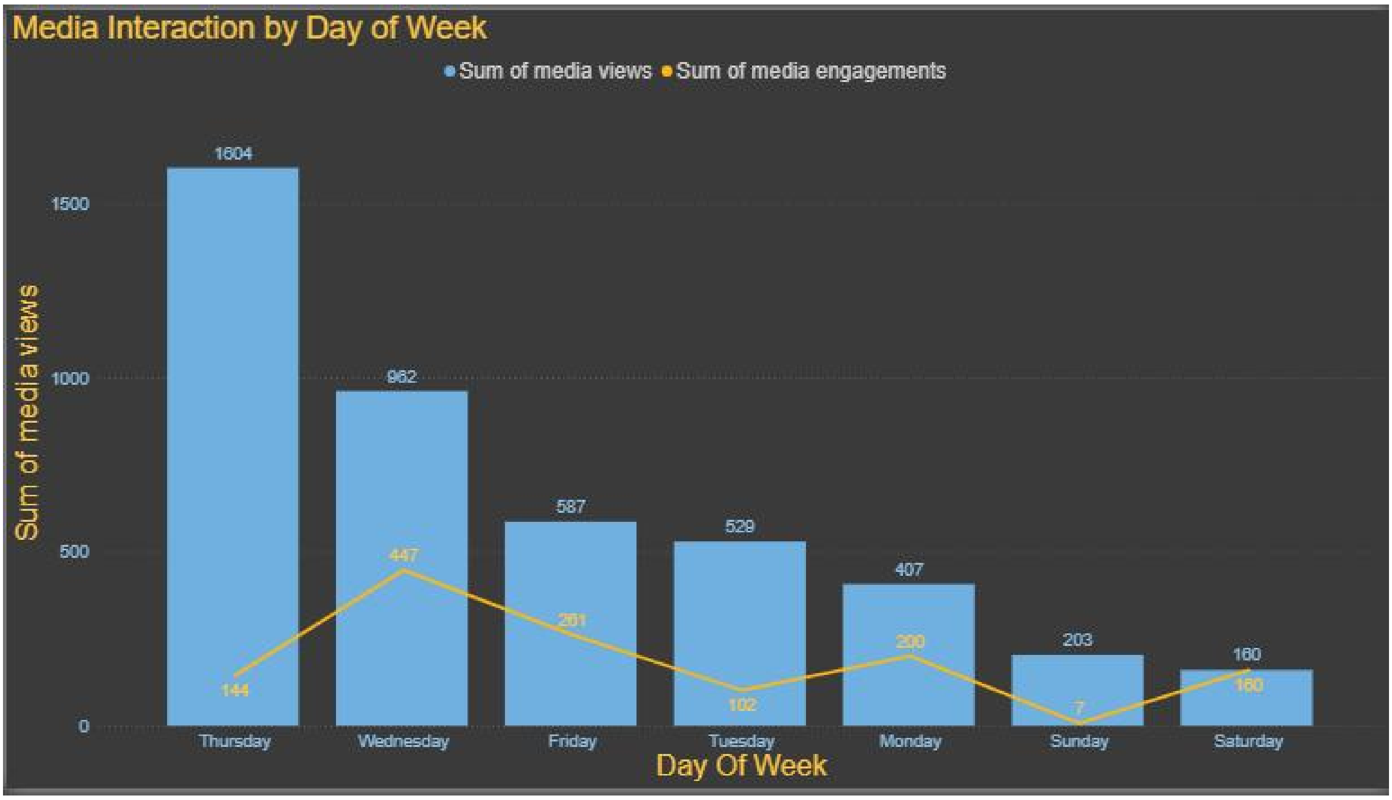
Compared the average engagement rate of tweets with app opens and without app opens. Applied filters including weekdays, posting time, impressions, tweet dates, character count, and tweet text conditions.

Note for Task 2: All the required filters specified in the assignment were implemented successfully, including weekday, posting time, even impressions, odd tweet date, character count greater than 30, exclusion of tweets containing the letter 'D', and the time-based visibility condition. After applying all filters simultaneously, the dataset did not contain any records satisfying every condition. Therefore, the visualization appears blank. This outcome reflects the filtered dataset rather than an implementation error.

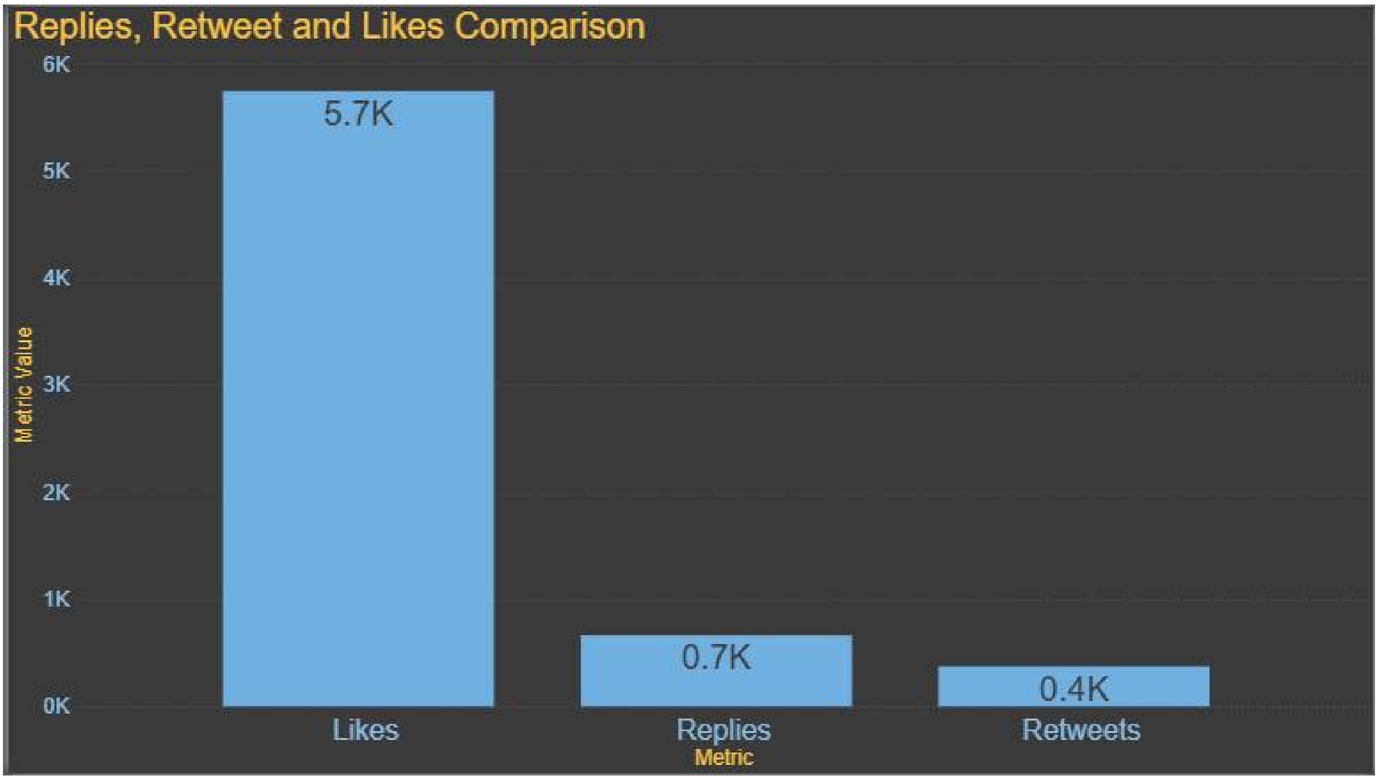


Task 3

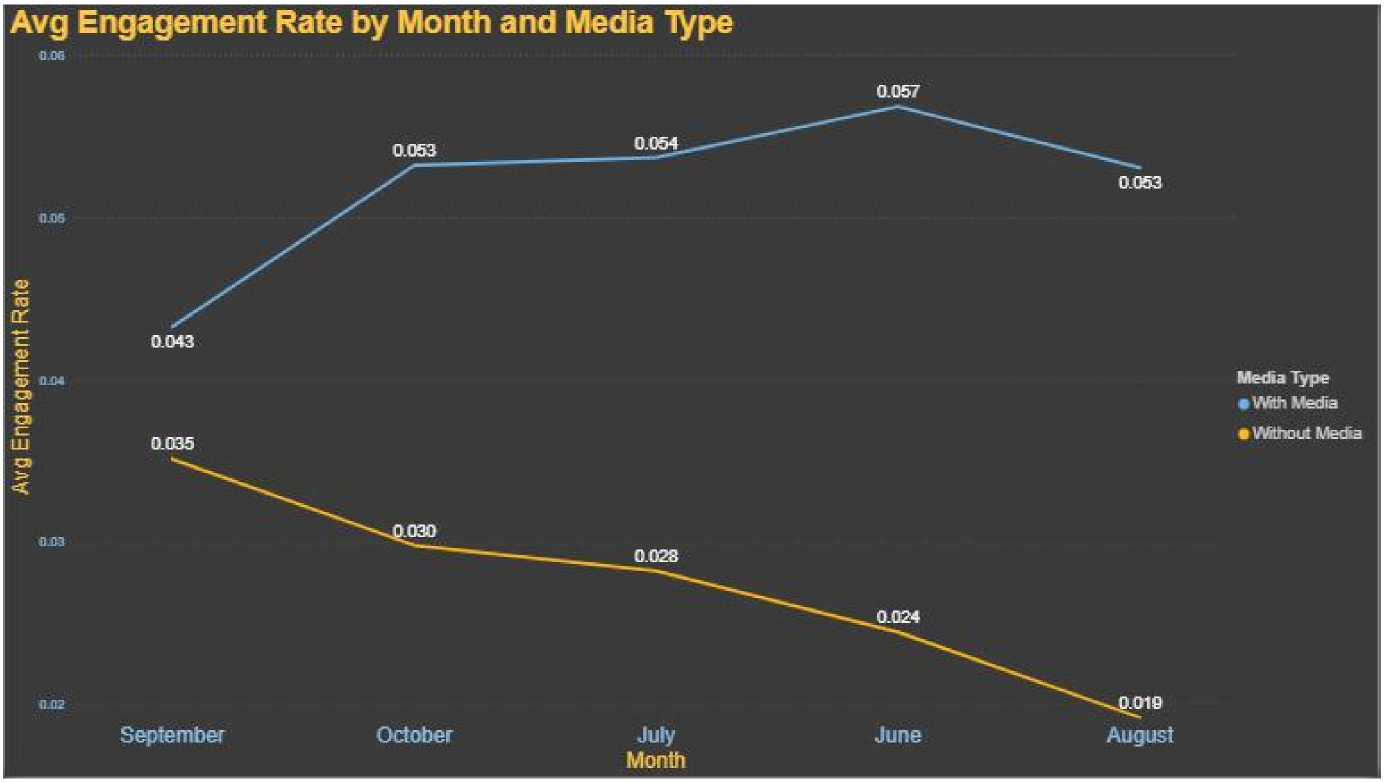
Developed a dual-axis visualization showing media views and media engagements by day of the week. Applied multiple filters and highlighted media interaction patterns.



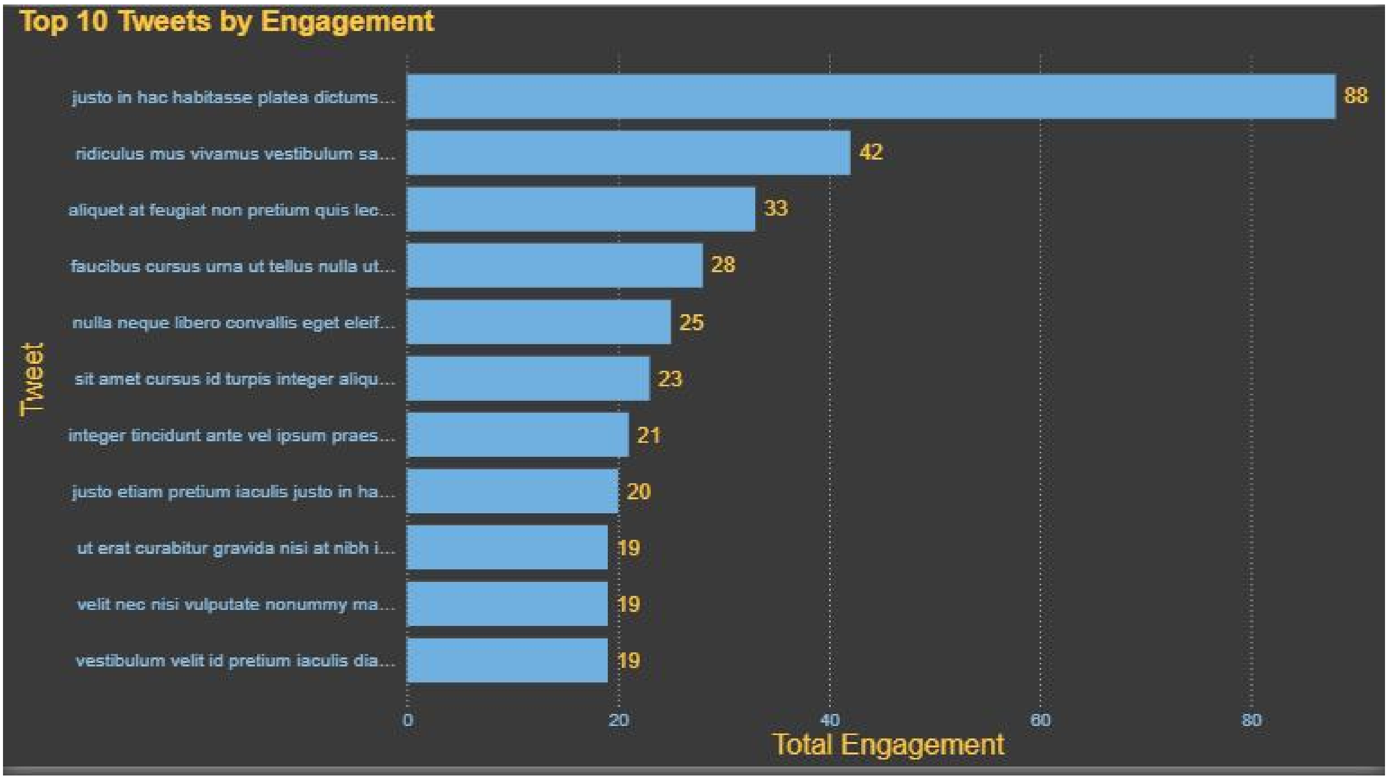
Task 4
Created a bar chart comparing replies, retweets, and likes for tweets posted between June and August 2020 using SUM aggregation.



Task 5
Built a line chart illustrating monthly average engagement rate trends for tweets with media and without media content.



Task 6
Developed a Top 10 Tweets by Engagement dashboard using total likes and retweets while applying the required dataset filters.



Skills and Competencies
During this internship, I strengthened the following technical and analytical skills:
- Microsoft Power BI
- Data Visualization
- Dashboard Design

- DAX Calculations
- Data Filtering
- Data Modeling
- Business Analytics
- Social Media Data Analysis
- Analytical Thinking
- Problem Solving

Challenges and Solutions

While completing the internship tasks, I encountered several challenges related to applying multiple filters simultaneously. In some cases, the filtered dataset contained very few or no matching records because of the strict assignment conditions. I carefully validated the dataset, created helper columns and measures using DAX, and ensured that all required filtering conditions were implemented correctly. This process improved my debugging and analytical skills while working with real-world data.

Outcomes and Impact

This internship provided practical exposure to business intelligence and dashboard development. By completing the assigned tasks, I gained confidence in using Power BI for creating interactive reports, performing data analysis, and presenting insights visually. The internship also improved my understanding of data-driven decision-making and strengthened my problem-solving abilities.

Conclusion

The internship was a valuable learning experience that enhanced both my technical knowledge and analytical skills. Working on Twitter analytics dashboards allowed me to apply theoretical concepts in a practical environment. I developed a deeper understanding of Power BI, DAX, and data visualization techniques, which will help me in future academic projects and professional opportunities in the field of Data Analytics.